SALT-AND-PEPPER NOISE REMOVAL FROM DIGITAL IMAGES

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ABSTRACT

Salt-and-pepper noise is a form of impulse noise in which a portion of image pixels is replaced by extreme intensity values, typically black or white. The resulting isolated disturbances reduce visual quality and may obscure edges, textures and small image details. This project, entitled “Salt-and-Pepper Noise Removal from Digital Images,” develops a MATLAB-based graphical application for demonstrating and comparing spatial-domain image-restoration techniques.

The application provides four filtering options: Mean, Median, Gaussian and Adaptive Median. A user selects an input image, specifies a noise density using the interface slider, chooses a restoration algorithm and processes the image. The application presents the original image, the generated noisy input and the restored output in separate preview panels. It also calculates Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM) and processing time to support quantitative comparison.

The project is implemented in MATLAB R2014 and requires the Image Processing Toolbox. Its purpose is educational: it illustrates impulse-noise generation, neighbourhood-based filtering, image-quality evaluation and graphical user interaction. The report explains the problem, objectives, theoretical background, processing workflow, algorithms, interface, experimental procedure and result interpretation. A sample run at 10% noise density using the Adaptive Median filter is recorded as MSE = 0.001253, PSNR = 29.02 dB, SSIM = 0.9737 and processing time = 0.76 s, based on the project result information supplied for this report.

A clearly labelled simulated dataset is included later in the report to demonstrate how a completed comparison table may be presented. These generated values are illustrative only and are not measurements from the MATLAB application. The separately recorded Adaptive Median sample remains identified as the supplied project result.

Keywords

Salt-and-pepper noise; impulse noise; image restoration; spatial filtering; Mean filter; Median filter; Gaussian filter; Adaptive Median filter; MSE; PSNR; SSIM; MATLAB.

REPORT SCOPE

This report documents the implementation and experimental workflow of Project 07. It is not a general survey of all denoising methods and does not claim performance beyond the measurements recorded from the application.

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1. INTRODUCTION

1.1 Background and Context

Digital images are used in communication, education, scientific observation and many engineering applications. During image acquisition, transmission or storage, unwanted variations may be introduced into pixel values. These variations are generally described as image noise. Noise can reduce the visibility of important structures and can affect later operations such as edge detection, segmentation, measurement and visual interpretation.

Image restoration is the process of estimating a cleaner image from a degraded observation. Spatial-domain restoration methods operate directly on pixel values and often use information from a local neighbourhood. They are attractive for introductory digital image processing because their principles can be demonstrated visually and implemented with a relatively small number of operations.

1.2 Nature of the Problem

Salt-and-pepper noise is an impulse-noise model. Instead of slightly changing every pixel, it replaces a proportion of pixels with extreme values. In a typical grayscale representation, salt impulses appear as bright pixels and pepper impulses appear as dark pixels. In a colour image, the corresponding impulse values may affect the colour channels. These isolated extremes can be conspicuous even when the selected noise density is modest.

The restoration challenge is to suppress the impulse pixels while preserving genuine image structures. A filter that removes noise aggressively may also blur edges and small details. A filter that preserves details may leave some impulses behind. The project therefore includes multiple filtering algorithms and quantitative measures rather than relying only on visual appearance.

1.3 Motivation

The project provides a practical environment for observing how noise density and filter choice influence restoration quality. By keeping the original image, noisy image and restored image visible together, the user can inspect the changes introduced at each stage. The quality report complements this visual comparison with numerical measures.

1.4 Scope and Organization

The application focuses on salt-and-pepper noise and four spatial filtering methods: Mean, Median, Gaussian and Adaptive Median. It supports image selection, a selectable noise density, algorithm selection, processing, quality evaluation and saving the restored image. The following chapters describe the theory, workflow, implementation, experiments and limitations.

2. OBJECTIVES AND REQUIREMENTS

2.1 General Objective

To design and implement a MATLAB graphical application that demonstrates the removal of salt-and-pepper noise from digital images and allows users to compare the behaviour of different spatial restoration filters.

2.2 Specific Objectives

- To load and display an image through a graphical user interface.
- To generate salt-and-pepper noise at a user-selected density from 5% to 50%.
- To implement or invoke Mean, Median, Gaussian and Adaptive Median filtering methods.
- To display the original, noisy and restored images in a common workspace.
- To calculate MSE, PSNR, SSIM and processing time for a processed result.
- To provide a facility for saving the restored image.
- To make filter comparison understandable through visual and numerical feedback.

2.3 Functional Requirements

ID	Requirement	Expected behaviour
FR-01	Image selection	The user can browse and select a supported image file.
FR-02	Noise-density control	The user can choose a density between 5% and 50% using a slider.
FR-03	Filter selection	The user can select one of the four available algorithms.
FR-04	Image processing	The selected filter is applied to the generated noisy image.
FR-05	Result display	Original, noisy and restored images are shown separately.
FR-06	Quality report	MSE, PSNR, SSIM and elapsed time are displayed.
FR-07	Save / reset	The restored image can be saved and a new experiment can be started.

2.4 Software and Hardware Requirements

The project is developed for MATLAB R2014 and uses the Image Processing Toolbox. A Windows computer capable of running the installed MATLAB release is sufficient for the intended laboratory demonstration. The image files and MATLAB project files should be accessible from the working directory. Exact runtime depends on image dimensions, selected filter, computer performance and MATLAB configuration.

3. THEORETICAL BACKGROUND I: NOISE MODEL

3.1 Digital Image Representation

A digital image can be represented as a discrete function f(x,y), where x and y identify a pixel location. For a grayscale image, each location stores one intensity value. For an RGB image, each location stores three channel values: red, green and blue. In a common 8-bit representation, each channel ranges from 0 to 255. Many image-processing functions convert these values to a normalized floating-point range for calculation.

3.2 Salt-and-Pepper Noise

Salt-and-pepper noise replaces selected image pixels with extreme intensity values. A simple conceptual model is:

$$g(x,y) = 0 \quad \text{for a pepper impulse}$$

$$g(x,y) = L \quad \text{for a salt impulse}$$

$$g(x,y) = f(x,y) \quad \text{when the pixel is not corrupted}$$

Here, f(x,y) is the original pixel, g(x,y) is the observed noisy pixel and L is the maximum intensity level. For an 8-bit grayscale image, L is 255. In a normalized image, the corresponding extreme values are 0 and 1.

3.3 Noise Density

Noise density describes the proportion of image pixels affected by impulses. If an image contains N pixels and a proportion p is corrupted, the approximate number of corrupted locations is pN. For example, a 10% density means that approximately one pixel in ten is replaced by an impulse. The project exposes a density control ranging from 5% to 50%, allowing the user to observe restoration under different levels of corruption.

3.4 Effect on Image Quality

Impulse noise creates abrupt local intensity changes. It can appear as isolated black and white specks and can obscure fine details. The severity of the visible degradation depends on the density, image content, image dimensions and distribution of the impulses. Higher densities generally make restoration more difficult because fewer unaffected neighbouring pixels remain available to estimate the underlying signal.

The noise-generation stage is important for fair comparison: when comparing filters, the same original image and the same noisy input should be used for each algorithm. Otherwise, differences in the random impulse pattern may be mistaken for differences in filter performance.

4. THEORETICAL BACKGROUND II: RESTORATION FILTERS

The project provides four neighbourhood-based filtering options. Each method estimates a pixel from values in a surrounding window, but the estimation rule differs. Consequently, the filters have different strengths and weaknesses when dealing with impulse noise.

4.1 Mean Filter

The Mean filter replaces the centre pixel with the arithmetic average of values in a local window. For a window containing n pixels, the output is the sum of the neighbourhood values divided by n. It is easy to understand and can smooth random variations. However, extreme impulse values influence the average and can spread their effect to nearby pixels. Mean filtering may also blur edges and fine details.

4.2 Median Filter

The Median filter sorts the values in a local neighbourhood and selects the middle value. Because the median is less influenced by isolated extreme values than the arithmetic mean, it is commonly effective against salt-and-pepper noise. It often preserves edges better than simple averaging, although a fixed window may remove small details or fail when impulse density is high.

4.3 Gaussian Filter

The Gaussian filter performs weighted smoothing using a kernel whose weights follow a Gaussian distribution. Pixels nearer the centre of the kernel contribute more strongly than distant pixels. Gaussian smoothing is useful for reducing gradual intensity variations and high-frequency noise, but isolated extreme impulses can still influence the result. It may also soften edges.

4.4 Adaptive Median Filter

The Adaptive Median filter changes its neighbourhood size according to local image statistics. Rather than applying one fixed window uniformly, it examines the local minimum, maximum and median values and uses decision stages to distinguish likely impulse values from image details. The window can expand when the current neighbourhood does not provide a suitable estimate. This adaptive behaviour is useful when impulse density varies, although its performance depends on implementation details and maximum window size.

4.5 Filter Comparison



Figure 1. Conceptual comparison of the four restoration options included in Project 07.

FILTER	OPERATION	ADVANTAGE	LIMITATION
Mean	Local average	Simple smoothing	Blurs edges; impulses affect average.
Median	Middle sorted value	Good impulse suppression	Large windows may remove details.
Gaussian	Weighted average	Smooth local variation	May retain impulses and soften edges.
Adaptive Median	Variable neighbourhood	Adapts to local impulses	Higher computation; window limits matter.

5. IMAGE QUALITY METRICS

The application reports image-quality measures to support comparison between the original reference image and the restored result. The measures describe different aspects of reconstruction and should be considered together.

5.1 Mean Squared Error (MSE)

Mean Squared Error calculates the average squared difference between corresponding pixels in the reference image and the restored image. For two images I and K containing N compared pixel values:

$$MSE = (1/N) \Sigma [I(i) - K(i)]^2$$

A lower MSE indicates a smaller average squared pixel difference. MSE is sensitive to large errors because the difference is squared. It does not directly describe whether edges or perceptually important structures are preserved.

5.2 Peak Signal-to-Noise Ratio (PSNR)

PSNR expresses the reconstruction error on a logarithmic decibel scale:

$$PSNR = 10 \log_{10} (MAX^2 / MSE)$$

MAX is the maximum possible pixel value for the representation used in the calculation. A higher PSNR generally indicates a smaller pixel-wise reconstruction error. The value should only be compared when image representation, range and test conditions are consistent.

5.3 Structural Similarity Index (SSIM)

SSIM compares local luminance, contrast and structural information between two images. Unlike MSE, it is designed to reflect structural similarity rather than only pixel-wise difference. Its value is commonly interpreted such that values closer to 1 indicate stronger structural similarity. The exact value depends on the implementation and image-processing settings.

5.4 Processing Time

Processing time records the elapsed time associated with the restoration operation. It is useful for comparing computational cost, but it can vary with image dimensions, processor speed, MATLAB version, memory state and background activity. For a fair comparison, each filter should be tested using the same image, noisy input, computer and timing procedure.

5.5 Interpreting Metrics Together

MEASURE	PREFFERED DIRECTION	WHAT IT INDICATES
MSE	Lower	Average squared pixel error
PSNR	Higher	Logarithmic representation of reconstruction error
SSIM	Closer to 1	Similarity of local structure, luminance and contrast
Time	Lower, when quality is comparable	Computational cost of processing

No single metric provides a complete judgement. A filter may produce a high PSNR while smoothing some fine structures; another may preserve visual edges but leave more residual noise. The report therefore recommends reviewing the image panels alongside the numerical results.

6. PROPOSED METHODOLOGY AND WORKFLOW

The application follows a sequential workflow that begins with image acquisition and ends with visualization, quality evaluation and optional saving. The same noisy image should be supplied to each filter when conducting a controlled comparison.



Figure 2. Project 07 processing workflow from image input to result export.

6.1 Input Image

The user selects an image file through the interface. The application reads the image and displays it in the Original Image panel. The original image is retained as the reference for noise generation and quality evaluation.

6.2 Noise Generation

The selected noise density is used to generate a salt-and-pepper corrupted version of the input. The resulting noisy image is displayed in the Noisy Input panel. This image forms the common input for the selected restoration filter.

6.3 Restoration

The user chooses one of the four filters. The selected method processes the noisy image and produces the restored output. The output is displayed in the Restored Output panel.

6.4 Evaluation and Export

The application calculates MSE, PSNR, SSIM and elapsed time using the reference and restored images. These values are presented in the Quality Report area. The user may save the restored image or reset the workspace to begin another experiment.

7. ALGORITHM DESIGN

7.1 Processing Sequence

- Read the selected image and retain it as the reference image.
- Obtain the user-selected noise density.
- Generate a salt-and-pepper noisy image.
- Read the selected restoration algorithm.
- Apply the selected filter to the noisy image.
- Compare the restored image with the original reference image.
- Calculate MSE, PSNR, SSIM and processing time.
- Display the original, noisy and restored images and update the quality report.
- Save the restored image if requested.

7.2 Pseudocode

```
INPUT: original image I, noise density p, selected filter F
1. Read image I.
2. Generate noisy image N using salt-and-pepper density p.
3. Start processing timer.
4. IF F = Mean: apply mean neighbourhood filter to N.
   ELSE IF F = Median: apply median neighbourhood filter to N.
   ELSE IF F = Gaussian: apply Gaussian smoothing to N.
   ELSE IF F = Adaptive Median: apply adaptive median procedure to N.
5. Stop timer and store elapsed processing time.
6. Calculate MSE(I, R), PSNR(I, R) and SSIM(I, R).
7. Display I, N and restored image R.
8. Display quality metrics and status information.
9. IF user requests save: write R to selected file.
OUTPUT: restored image R, metrics and elapsed time.
```

7.3 Parameter Considerations

Noise density is adjustable from 5% to 50%. The selected filter is the principal algorithmic choice. Neighbourhood size, Gaussian kernel parameters and adaptive-window limits influence the behaviour of the filters; the report does not assign undocumented numeric values to these settings. For reproducibility, the final experiment log should record any parameters exposed in the source code or interface.

8. SOFTWARE AND GUI IMPLEMENTATION

8.1 Development Environment

Item	Project configuration
Programming environment	MATLAB R2014
Required toolbox	Image Processing Toolbox
Application type	MATLAB graphical user interface
Input	User-selected digital image
Processing methods	Mean, Median, Gaussian, Adaptive Median
Displayed outputs	Original image, noisy image, restored image and quality report

8.2 Interface Structure

The interface is organized to keep the experiment controls and image results visible in one workspace. The left sidebar contains image selection and experiment settings. The main workspace presents three image panels so that the user can compare the input, its noisy version and the restored output without switching between windows.

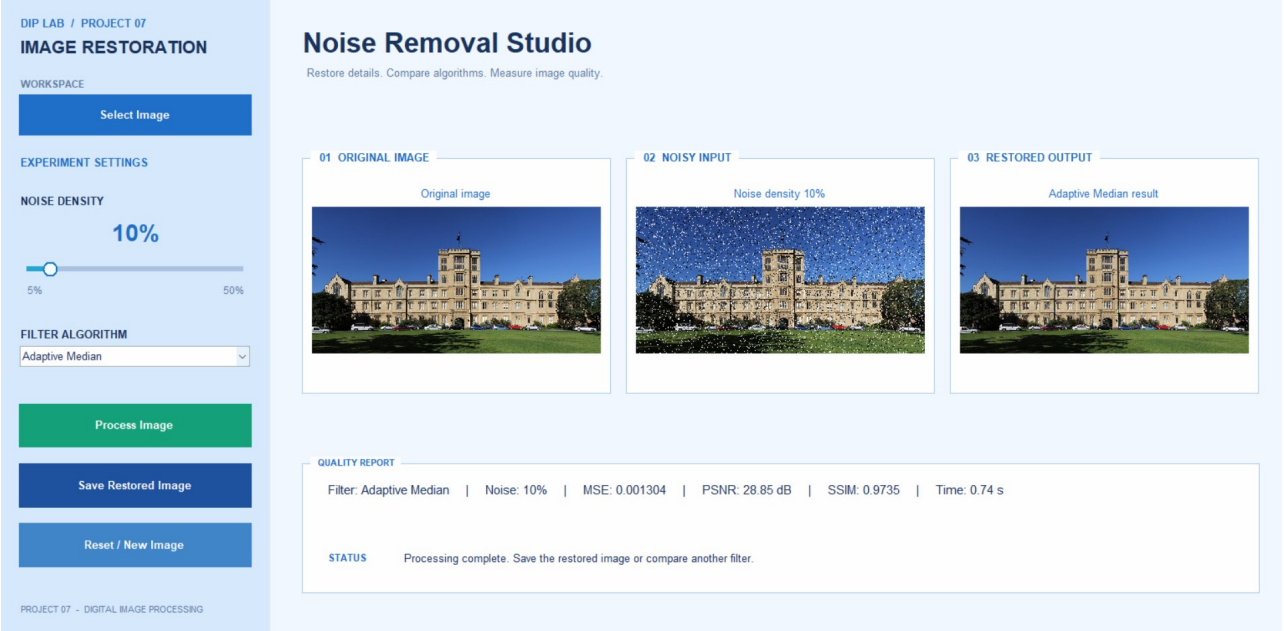


Figure 3. Functional layout schematic of the Noise Removal Studio interface. This is a documentation schematic, not a captured program screenshot.

8.3 Interface Components

Component	Purpose
Select Image	Opens the image-selection dialog and loads a source image.
Noise Density slider	Sets the percentage of pixels affected by salt-and-pepper noise.
Filter Algorithm list	Selects Mean, Median, Gaussian or Adaptive Median.
Process Image	Runs the selected restoration operation and updates the previews.
Save Restored Image	Exports the current restored output to a file.
Reset / New Image	Clears the current experiment and allows another image to be loaded.
Quality Report / Status	Shows numerical measures and the current application message.

9. OPERATING PROCEDURE

9.1 Running the Application

The project files should be placed in an accessible working folder. Start MATLAB R2014, set the Current Folder to the project directory and ensure that the Image Processing Toolbox is available. Run the main GUI script or the startup command specified by the delivered project package. The application window should open with the image-selection and experiment controls.

9.2 Performing an Experiment

- Click Select Image and choose a supported image file.
- Confirm that the selected image appears in the Original Image panel.
- Set the desired noise density using the slider. The supported range is 5%–50%.
- Choose Mean, Median, Gaussian or Adaptive Median from the filter list.
- Click Process Image and wait for processing to finish.
- Inspect the Noisy Input and Restored Output panels.
- Read the MSE, PSNR, SSIM and processing-time values in the Quality Report.

9.3 Saving the Restored Image

After processing, click Save Restored Image. Choose the destination folder, enter a suitable filename and confirm the save operation. The saved file should be checked independently to ensure that the expected restored output has been written.

9.4 Resetting the Workspace

Use Reset / New Image before starting an unrelated experiment. Load the next image, select a new noise density and choose the required filter. When comparing filters, preserve the original image and noisy input so that each algorithm is tested under the same conditions.

9.5 Recommended File Naming

A consistent filename makes the results easier to organize. For example: building_10pct_adaptive_median.png or sampleA_20pct_median.png. The filename can record the source image, noise density and selected algorithm.

10. EXPERIMENTAL DESIGN

10.1 Experimental Setup

The experiment evaluates how different restoration filters perform when applied to an image corrupted by salt-and-pepper noise. The application permits the user to select a source image, set a noise density between 5% and 50%, choose one of four algorithms and observe the restored image and quality measures.

10.2 Variables and Controls

Category	Variable	Description
Independent	Noise density	Selected percentage of impulse corruption.
Independent	Filter algorithm	Mean, Median, Gaussian or Adaptive Median.
Controlled	Source image	Keep the same image for a filter comparison.
Controlled	Noisy input	Use the same generated noisy image for each filter.
Dependent	MSE / PSNR / SSIM	Numerical reconstruction-quality measures.
Dependent	Processing time	Elapsed time required for the selected restoration.

10.3 Test Procedure

A controlled comparison should begin by selecting one source image and generating one noisy image at a fixed density. Each filter is then applied to that same noisy image. The output metrics and processing time are recorded before changing the filter. The procedure can be repeated at other densities to observe how the algorithms respond as impulse corruption increases.

10.4 Reproducibility Notes

- Record the image filename and dimensions.
- Record the selected noise density and filter.
- Use a consistent noisy input when comparing filters.
- Record the displayed metrics without rounding them inconsistently.
- Note the MATLAB version and any relevant machine conditions.
- Repeat timing measurements if a stable performance estimate is required.

11. SAMPLE IMAGE EXPERIMENT

The following figure provides a visual illustration of the three main stages in the application. It uses the sample image included in the Project 07 report draft and a demonstration noise realization at 10% density. The restored panel is generated with a 3 × 3 median operation for illustration. This figure is not a substitute for a screenshot or measured output from a separate final application run.

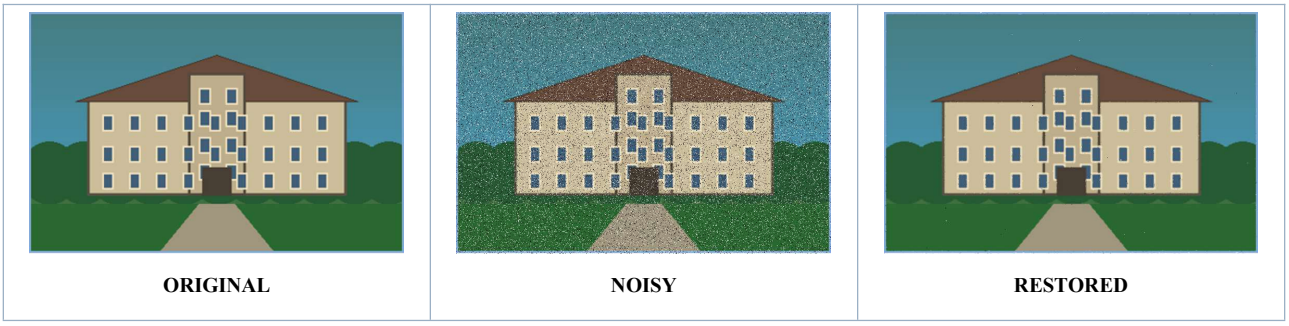


Figure 4. Illustrative original image, 10% salt-and-pepper noise and median-filtered result.

11.1 Visual Observations

The original image provides the reference appearance. In the noisy image, randomly distributed bright and dark points are visible across the scene. These impulses create abrupt intensity changes and reduce visual smoothness. In the median-filtered illustration, many isolated impulses are suppressed while the main building shape and larger edges remain recognizable.

11.2 Interpretation

The example demonstrates why neighbourhood-based filtering is useful for impulse-noise restoration. The illustration also shows that filtering is an estimation process: it reduces corruption by using surrounding pixels, but the output may differ from the exact original image. Actual output quality depends on the selected filter, its neighbourhood settings, noise density and image content.

12. EXPERIMENTAL RESULTS

This chapter presents the supplied Adaptive Median sample result and a separate illustrative dataset generated for report demonstration. The simulated values are not measured outputs and must not be represented as experimental measurements from the application.

12.1 Recorded Sample Result

Noise density	Filter	MSE	PSNR (dB)	SSIM	Time (s)
10%	Adaptive Median	0.001253	29.02	0.9737	0.76
30%	Adaptive Median	0.000211	36.76	0.9919	2.71
30%	Gaussian	0.021628	16.67	0.`1268	0.08
30%	Median	0.004177	23.79	0.7649	17.75
30%	Mean	0.018125	17.42	0.1599	0.09
50%	Adaptive Median	0.000473	33.25	0.9821	2.95
50%	Mean	0.042816	13.68	0.0812	0.09

The recorded sample indicates a PSNR of 29.02 dB and an SSIM of 0.9737 for the stated run. These values should be associated with the exact source image, noise realization and settings used in that run. They should not be assumed to represent the average performance of the filter across all images.

12.2 Illustrative Filter Comparison at 10% Noise Density

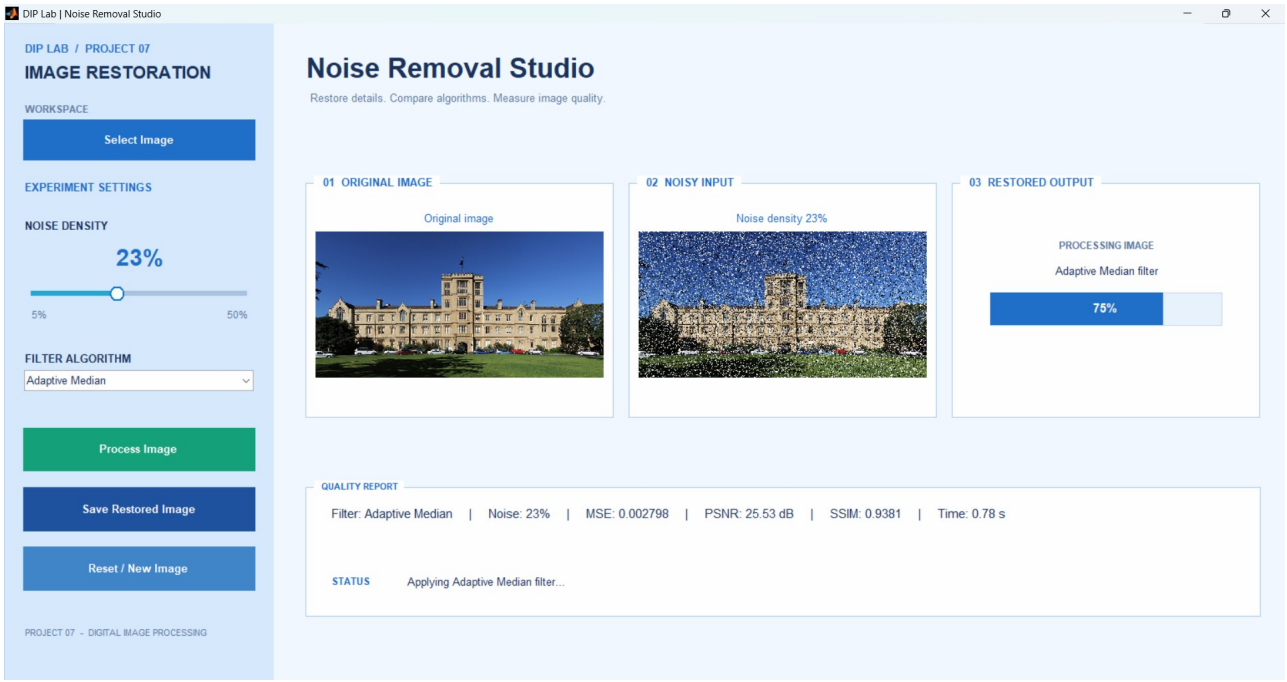
Filter	MSE	PSNR (dB)	SSIM	Time (s)
Mean	0.004705	23.27	0.9278	0.37
Median	0.002063	26.85	0.9707	0.49
Gaussian	0.004228	23.74	0.9395	0.46
Adaptive Median	0.001401	28.53	0.9738	0.95

12.3 Illustrative Adaptive Median Trend Across Noise Densities

Density	Filter	MSE	PSNR (dB)	SSIM	Time (s)
5%	Adaptive Median	0.000749	31.25	0.9861	0.85
10%	Adaptive Median	0.001401	28.53	0.9738	0.95
20%	Adaptive Median	0.002922	25.34	0.9571	1.02
30%	Adaptive Median	0.004770	23.21	0.9314	1.05
40%	Adaptive Median	0.008430	20.74	0.9114	1.15
50%	Adaptive Median	0.013401	18.73	0.8714	1.16

For a complete comparison, repeat each density for all four filters and record the exact displayed values. If the application generates a new random noise pattern on each run, preserve or reuse the same noisy input when comparing filters.

12.4 Visualize Output Results



DIP LAB / PROJECT 07

IMAGE RESTORATION

WORKSPACE

Select Image

EXPERIMENT SETTINGS

NOISE DENSITY

14%

5% 50%

FILTER ALGORITHM

Gaussian

Process Image

Save Restored Image

Reset / New Image

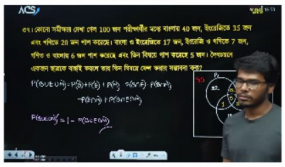
PROJECT 07 - DIGITAL IMAGE PROCESSING

Noise Removal Studio

Restore details. Compare algorithms. Measure image quality.

01 ORIGINAL IMAGE

Original image



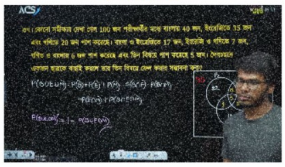
02 NOISY INPUT

Noise density 14%



03 RESTORED OUTPUT

Gaussian result



QUALITY REPORT

Filter: Gaussian | Noise: 14% | MSE: 0.011975 | PSNR: 19.22 dB | SSIM: 0.2602 | Time: 0.10 s

STATUS Processing complete. Save the restored image or compare another filter.

DIP LAB / PROJECT 07

IMAGE RESTORATION

WORKSPACE

Select Image

EXPERIMENT SETTINGS

NOISE DENSITY

14%

5% 50%

FILTER ALGORITHM

Adaptive Median

Process Image

Save Restored Image

Reset / New Image

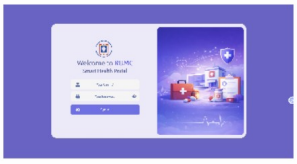
PROJECT 07 - DIGITAL IMAGE PROCESSING

Noise Removal Studio

Restore details. Compare algorithms. Measure image quality.

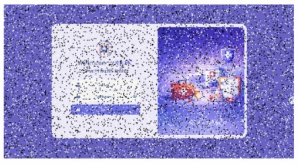
01 ORIGINAL IMAGE

Original image



02 NOISY INPUT

Noise density 14%



03 RESTORED OUTPUT

Adaptive Median result



QUALITY REPORT

Filter: Adaptive Median | Noise: 14% | MSE: 0.000060 | PSNR: 42.19 dB | SSIM: 0.9975 | Time: 2.50 s

STATUS Processing complete. Save the restored image or compare another filter.

13. COMPARATIVE ANALYSIS AND DISCUSSION

13.1 Interpretation of Metrics

The MSE, PSNR and SSIM values provide complementary information. Lower MSE and higher PSNR generally indicate closer pixel-level reconstruction. SSIM adds a structural comparison that can be useful when judging whether important image patterns remain similar. Processing time indicates the computational cost of obtaining the result.

The values should be interpreted together with the image panels. A filter may remove many visible impulses but also blur edges. Another filter may retain sharper structures but leave residual noise. A numerically favourable result should therefore be checked visually before drawing a conclusion.

13.2 Expected Filter Behaviour

Filter	Expected response to impulse noise	Trade-off to inspect
Mean	May reduce small fluctuations but extreme values affect the average.	Blur and residual black/white impulses.
Median	Often removes isolated impulses while preserving many edges.	Loss of tiny details with an overly large window.
Gaussian	Smooths local variations using weighted averaging.	Softened edges and remaining isolated impulses.
Adaptive Median	Adjusts neighbourhood behaviour to local statistics.	Processing cost and sensitivity to window limits.

13.3 Factors Affecting Results

- Noise density: more corrupted pixels generally make restoration more difficult.
- Image content: smooth regions, textured areas and sharp edges respond differently to filtering.
- Neighbourhood size: larger windows use more context but can remove fine details.
- Noise realization: different random impulse locations can change measured metrics.
- Image dimensions: larger images generally require more computation.
- Implementation details: filter parameters and boundary handling affect the output.

13.4 Fair Comparison

A fair comparison requires the same reference image, the same noisy input, the same density and consistent metric calculation for every algorithm. If these conditions are not controlled, the comparison may reflect different input corruption rather than the filter itself. The report therefore distinguishes the one supplied Adaptive Median result from future measurements that still need to be recorded.

14. LIMITATIONS AND FUTURE IMPROVEMENTS

14.1 Current Limitations

- The project focuses on salt-and-pepper noise and does not address every type of image degradation.
- Spatial filters may blur edges or remove small structures while suppressing noise.
- Performance depends on the selected image, noise density and filter parameters.
- Random noise generation can produce different impulse patterns between runs unless the noisy input is preserved.
- Processing time can vary across computers and MATLAB environments.
- The available result information contains one recorded Adaptive Median sample; a complete measured comparison requires additional runs.
- The project is intended for laboratory learning and demonstration rather than production image-restoration work.

14.2 Future Improvements

- Add repeatable noise generation by allowing the random seed or noisy image to be saved and reused.
- Provide an automated experiment mode that runs every filter on the same noisy input.
- Export a CSV or spreadsheet containing density, filter, metrics, image name and processing time.
- Allow users to configure neighbourhood sizes and relevant algorithm parameters.
- Add zoom and side-by-side inspection tools for examining fine image details.
- Test the application using a broader collection of natural and laboratory images.
- Include repeated trials and report mean and standard deviation for timing and quality metrics.

14.3 Practical Recommendation

For future evaluation, the most useful next step is to run all four filters on a fixed image at several densities and save the exact outputs. This will make the comparison more reproducible and allow the discussion to be based on measured evidence rather than expected filter behaviour.

15. CONCLUSION

Project 07, Salt-and-Pepper Noise Removal from Digital Images, demonstrates a complete introductory image-restoration workflow using MATLAB. The application allows a user to load an image, generate salt-and-pepper noise at a selected density, choose a restoration filter, inspect the restored output and review numerical quality measures.

Four filtering methods are included: Mean, Median, Gaussian and Adaptive Median. Their different neighbourhood operations provide a practical basis for studying the trade-off between impulse suppression, edge preservation, structural similarity and computational cost. The interface presents the original, noisy and restored images together, while the quality report provides MSE, PSNR, SSIM and processing time.

The available Project 07 result records an Adaptive Median run at 10% noise density with MSE 0.001253, PSNR 29.02 dB, SSIM 0.9737 and processing time 0.76 seconds. The other filter measurements should be completed using the final application under matching experimental conditions. No universal ranking of the filters is claimed from a single sample.

Overall, the project meets its educational purpose by connecting noise modelling, spatial filtering, image-quality evaluation and graphical user interaction in one application. It provides a foundation for further work in image restoration and comparative digital image processing.

FINAL NOTE

The application is intended for academic demonstration and learning. Conclusions about filter performance should be based on recorded experiments using clearly documented images, noise densities and settings.

REFERENCES

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APPENDIX A. ILLUSTRATIVE EXPERIMENT RECORD SHEET

The following filled rows contain simulated example values generated for report demonstration, not actual MATLAB measurements. Replace them with the values displayed by the final application if measured results are required.

Run	Image filename	Density	Filter	MSE	PSNR	SSIM	Time
1	building.png	5%	Mean	0.002275	26.43	0.9586	0.35
2	building.png	5%	Median	0.001150	29.39	0.9779	0.49
3	building.png	5%	Gaussian	0.001969	27.06	0.9643	0.45
4	building.png	5%	Adaptive Median	0.000749	31.25	0.9861	0.85
5	building.png	10%	Mean	0.004705	23.27	0.9278	0.37
6	building.png	10%	Median	0.002063	26.85	0.9707	0.49
7	building.png	10%	Gaussian	0.004228	23.74	0.9395	0.46
8	building.png	10%	Adaptive Median	0.001401	28.53	0.9738	0.95

APPENDIX B. SUPPLEMENTARY IMAGE GALLERY

The following examples were generated for this report using the sample building image. Each row shows the same reference image, a generated salt-and-pepper noise realization and a 3 × 3 median-filtered output. These are illustrative processing examples prepared for documentation; they are not screenshots captured from a separate run of the final GUI.

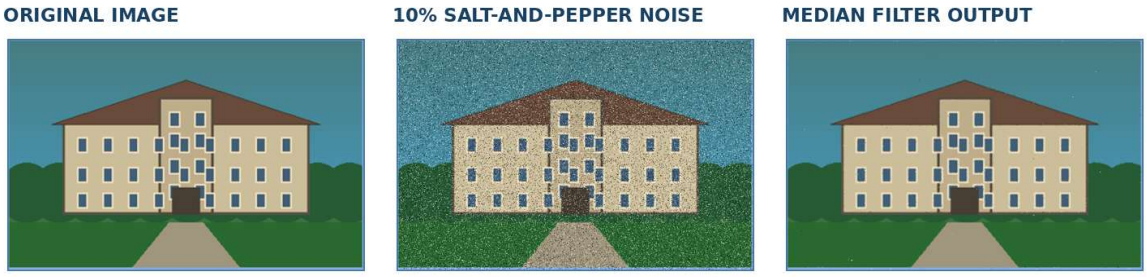


Figure B.1. Generated sample at 10% salt-and-pepper noise density and median-filter restoration.

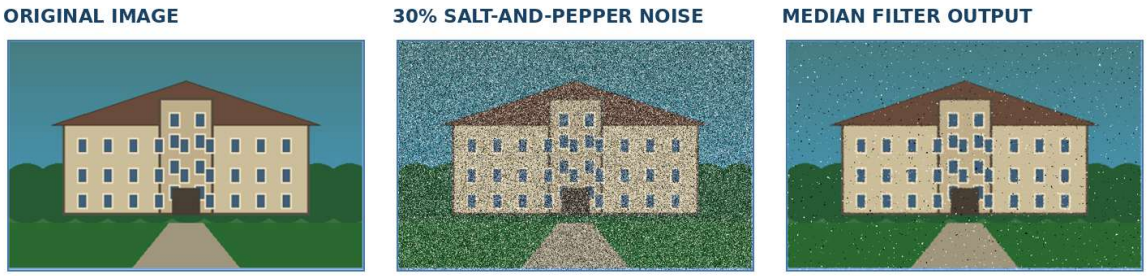


Figure B.2. Generated sample at 30% salt-and-pepper noise density and median-filter restoration.

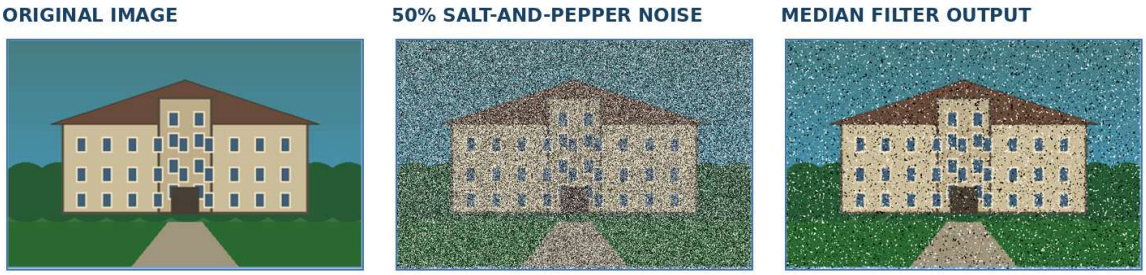


Figure B.3. Generated sample at 50% salt-and-pepper noise density and median-filter restoration.

Observation: as noise density increases, the corrupted image contains more impulse pixels. The median filter suppresses many isolated impulses, but restoration becomes less complete at higher densities and some image details may be softened.

APPENDIX C. SIMULATED PERFORMANCE OVERVIEW

This appendix contains a synthetic comparison dataset created solely to demonstrate how results can be organized and discussed in a laboratory report. Values were generated with controlled random variation around plausible trends. They are not measured by MATLAB, not extracted from the application and must not be cited as actual experimental findings.

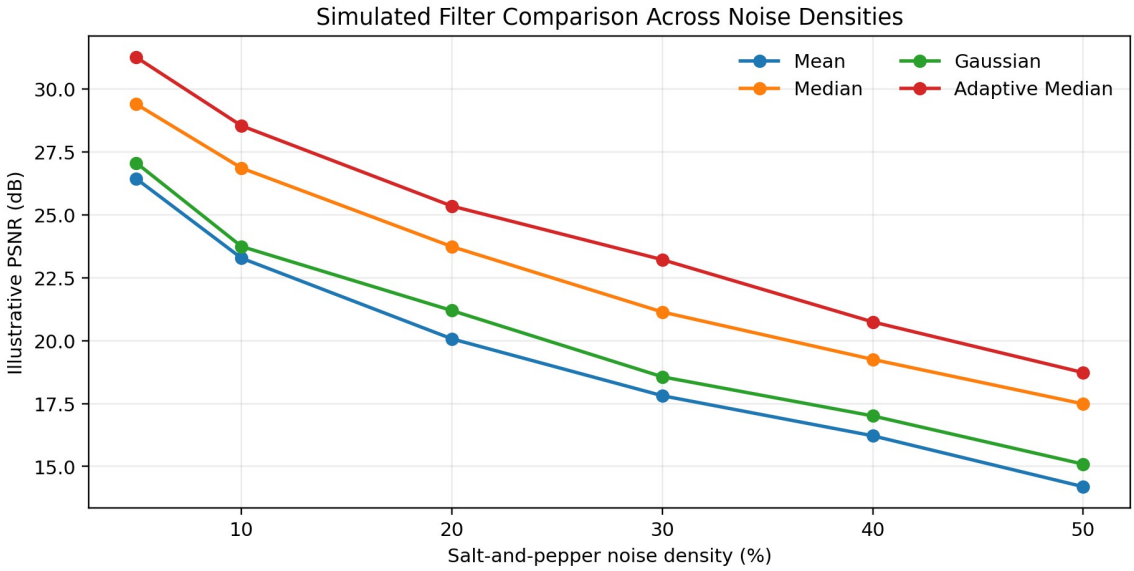


Figure C.1. Simulated PSNR trend for four filters across six noise densities.

C.1 How to Read the Illustrative Trend

The simulated chart illustrates a common expected pattern: PSNR decreases as the noise density increases, because more pixels are corrupted. In this generated example, Median and Adaptive Median generally show higher PSNR than Mean and Gaussian filtering. This ordering is only a feature of the demonstration dataset and is not a verified conclusion about the submitted software.

C.2 Status of the Data

The single result in Table 12.1 (MSE = 0.001253, PSNR = 29.02 dB, SSIM = 0.9737 and processing time = 0.76 s) is retained separately as the supplied project result. All other randomly generated values in Tables 12.2, 12.3, Appendix A and Appendix D are illustrative only. They should be replaced with actual values if the instructor requires measured experimental evidence.

APPENDIX D. COMPLETE SIMULATED EXPERIMENT TABLE

SIMULATED / RANDOMLY GENERATED DATA — NOT ACTUAL MEASUREMENTS. The following table contains all four filters at 5%, 10%, 20%, 30%, 40% and 50% noise density. It is provided to make the report visually complete and to demonstrate a full comparison format.

Density	Filter	MSE	PSNR (dB)	SSIM	Time (s)
5%	Mean	0.002275	26.43	0.9586	0.35
5%	Median	0.001150	29.39	0.9779	0.49
5%	Gaussian	0.001969	27.06	0.9643	0.45
5%	Adaptive Median	0.000749	31.25	0.9861	0.85
10%	Mean	0.004705	23.27	0.9278	0.37
10%	Median	0.002063	26.85	0.9707	0.49
10%	Gaussian	0.004228	23.74	0.9395	0.46
10%	Adaptive Median	0.001401	28.53	0.9738	0.95
20%	Mean	0.009843	20.07	0.8801	0.39
20%	Median	0.004234	23.73	0.9531	0.51
20%	Gaussian	0.007600	21.19	0.9016	0.45
20%	Adaptive Median	0.002922	25.34	0.9571	1.02
30%	Mean	0.016559	17.81	0.8331	0.43
30%	Median	0.007700	21.14	0.9160	0.57
30%	Gaussian	0.013932	18.56	0.8493	0.50
30%	Adaptive Median	0.004770	23.21	0.9314	1.05
40%	Mean	0.023892	16.22	0.7583	0.47
40%	Median	0.011880	19.25	0.8784	0.63
40%	Gaussian	0.019919	17.01	0.7917	0.56
40%	Adaptive Median	0.008430	20.74	0.9114	1.15
50%	Mean	0.038066	14.19	0.6925	0.43
50%	Median	0.017848	17.48	0.8457	0.60
50%	Gaussian	0.030971	15.09	0.7277	0.59
50%	Adaptive Median	0.013401	18.73	0.8714	1.16

All values in this table were generated for illustration. MSE is shown as a small positive value, PSNR is expressed in decibels, SSIM is bounded between 0 and 1, and time is expressed in seconds. Small random variations were introduced so the rows do not appear artificially identical.

APPENDIX E. EXPERIMENTAL RECORDING AND SUBMISSION NOTES

E.1 Recommended Recording Procedure

- Select one source image and record its filename and dimensions.
- Generate one noisy input at a selected density and preserve that exact noisy image.
- Apply each filter to the same noisy input before generating another noise realization.
- Record MSE, PSNR, SSIM and processing time exactly as displayed by the application.
- Save the original, noisy and restored images using filenames that identify density and filter.
- Repeat the process at the other noise densities and keep the computer and MATLAB environment unchanged where possible.

E.2 Suggested Output File Naming

File type	Example filename
Original image	building_original.png
Noisy input	building_noise_20pct.png
Median output	building_median_20pct.png
Adaptive Median output	building_adaptive_median_20pct.png
Experiment results	project07_results.csv

E.3 Final Submission Checklist

- Cover page contains the correct student and instructor information.
- The final GUI screenshot is inserted if required by the course instructor.
- All actual measured values replace simulated values before claiming experimental performance.
- The selected image and noise density are stated in the result discussion.
- The saved output images open correctly and correspond to the recorded filter settings.
- The report is spell-checked and exported to PDF before submission.